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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Graph Algorithms (course)

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Course outline

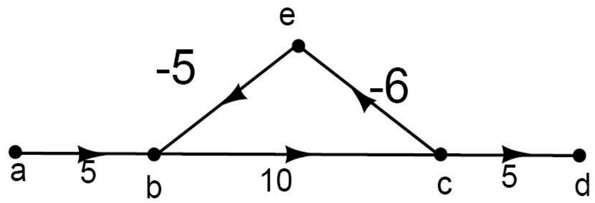
Week 2 Assignment 2

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

As per our records you have not submitted this assignment.

1) In the graph given, $\alpha(a,b)$= ***2 points***



- ∞
- $-\infty$
- 0
- 5

No, the answer is incorrect.
Score: 0

Accepted Answers:
$-\infty$

2) State True or False ***1 point***

For any directed graph *G*, $\alpha(u,v) \le \delta(u,v)$ for any pair of vertices *u* and *v* of *G*.

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
True

3) State True or False **1 point**

Let *G* be a graph with no negative cycles then *α(u,v)<δ(u,v)*

- True
- False

No, the answer is incorrect.
Score: 0

Accepted Answers:
False

4) Let *G=(V,∈,W,s)* be an instance of a SSSP Problem. Let *δ(v)* denote the weight of the shortest path from *s* to *v*. Write down Bellman Equation satisfied by values *δ(u),u∈V*. **3 points**

- $\delta(s)= 0\ and\ \delta(u)= \underset{v \neq u}{Min} \{\delta(v)+w(v,u)\}\ for\ u \neq s, u \in V$
- $\delta(s)= 1 and\ \delta(u)= \underset{v \neq u}{Min} \{\delta(v)+w(v,u)\}\ for\ u \neq s, u \in V$
- $\delta(s)= 0\ and\ \delta(u)= \underset{v = u}{Min} \{\delta(v)+w(v,u)\}\ for\ u \neq s, u \in V$
- $\delta(s)= 0\ and\ \delta(u)= \underset{v \neq u}{Min} \{\delta(v)+w(u,v)\}\ for\ u \neq s, u \in V$

No, the answer is incorrect.
Score: 0

Accepted Answers:
$\delta(s)= 0\ and\ \delta(u)= \underset{v \neq u}{Min} \{\delta(v)+w(v,u)\}\ for\ u \neq s, u \in V$